

Eric J. Earley

(He/him)

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Curriculum Vitae

Updated 2026-02-23

SUMMARY

I am an Assistant Research Professor with the Bone-Anchored Limb Research Group and the Department of Orthopedics at the University of Colorado Anschutz Medical Campus, and the Limb Restoration Program at University of Colorado Hospital. My research focuses on characterizing and augmenting the sensorimotor experiences of patients with amputation and the incorporation of their prosthesis into the cognitive representation of their body schema.

Understanding these afferent and efferent phenomena allows for the development of technologies and targeted therapies to strengthen prosthesis embodiment, alleviate phantom limb pain, and improve overall quality of life.

This requires a transdisciplinary and collaborative approach bringing together engineers, scientists, surgeons, clinicians, therapists, and patients to develop efficacious and novel technologies that can be impactful not only in a laboratory setting, but at home and during daily use.

EDUCATION

Postdoctoral Training	Chalmers University of Technology	2020 – 2023
Ph.D. Biomedical Engineering	Northwestern University	2014 – 2020
M.S. Biomedical Engineering	Northwestern University	2012 – 2014
B.S. Engineering: Mechanical Specialty	Colorado School of Mines	2008 – 2012

RESEARCH EXPERIENCE

Assistant Research Professor

2023 – present

Bone-Anchored Limb Research Group ♦ University of Colorado

Department of Orthopedics ♦ University of Colorado Anschutz Medical Campus

Limb Restoration Program ♦ University of Colorado Hospital

Facilitated an international consensus study amongst prosthetists to establish recommendations for prosthetic componentry for patients with bone-anchored limbs. Developing an amputation-specific pain survey to characterize subtypes of pain – phantom, residual, bone, muscle, nerve, back, shoulder, and neck – experienced by patients with amputation. Proposing a novel framework for osseophenomena – uniquely enhanced somatosensory and neuropsychological experiences arising from direct skeletal attachment of prosthetic limbs. Overseeing IDE clinical trials and coordinating FDA Expanded Access requests for osseointegrated implants.

Research estimated to yield at least six first-author and one middle-author peer-reviewed journal articles, two first-author and one middle-author international conference papers, and seven first-author international conference abstracts to date.

Postdoctoral Research

2020 - 2023

Bionic Arms Area ♦ Center for Bionics and Pain Research

Department of Electrical Engineering ♦ Chalmers University of Technology

Mentor: Prof. Max Ortiz-Catalán

Oversaw clinical trials providing and upgrading neuromusculoskeletal implants for transradial prostheses. Proposed methods of characterizing perceptions elicited via varied neurostimulation modulations. Developed signal processing algorithms to remove neurostimulation artifacts from implanted EMG sensors and mathematical models to probe electrical connectivity between implanted sensors. Investigated longitudinal behavioral and functional impacts of prosthesis use at home. Mentored three PhD students and seven Master's students during their theses.

Research yielded at least eighteen peer-reviewed publications to date: seven first-author, one last-author, and seven middle-author journal articles, two first-author and one middle-author international conference papers.

Doctoral Research

2014 – 2020

Center for Bionic Medicine ♦ Shirley Ryan AbilityLab

Department of Biomedical Engineering ♦ Northwestern University

Thesis Title: *Sensory Substitution in the Presence of Vision: Providing Joint Speed Feedback to Improve Myoelectric Prosthesis Control and Adaptation.* [Dissertation.](#)

Advisors: Dr. Levi Hargrove, Prof. Jon Sensinger

Used sensory feedback to improve motor adaptation by providing information not accurately available via vision, as determined by psychophysical analysis.

Research yielded five peer-reviewed first-author publications: three journal articles and two conference papers.

Master's Research

2012 – 2014

Center for Bionic Medicine ♦ Rehabilitation Institute of Chicago

Department of Biomedical Engineering ♦ Northwestern University

Advisor: Dr. Levi Hargrove

Improved control of partial-hand prostheses through optimization of EMG pattern-recognition parameters and dynamic window lengths while preserving wrist mobility.

Research yielded three peer-reviewed first-author publications: one journal article and two conference papers.

Research Internship

2012

BioMechatronics Development Laboratory

Department of Bioengineering ♦ University of Colorado Anschutz Medical Campus

Advisor: Dr. Richard F. Weir

Designed thumb actuation mechanism and housing and created SolidWorks models of prototype 3-DOF prosthetic hand.

PUBLICATIONS

Peer-Reviewed Journal Articles

1. **Earley EJ**, Shaw KG, Awad ME, Gaffney BMM, Christiansen CL, Iorio ML, Stoneback JW, Melton DH. "Phantom Limb, Residual Limb, and Bodily Pains Relate Differently to Perceived Overall Pain, Quality of Life, and Activity Level in Persons with Unilateral Lower Limb Amputation," *Physical Medicine & Rehabilitation*, 2026. [Links](#).
2. **Earley EJ**, Milius DW, Awad ME, Melton DH, Ahmed K, Leijendekkers R, Potter BK, Stevens PM, Gaffney BMM, Christiansen CL, Stoneback JW. "Establishing Consensus for Prescription of Prosthetic Components for Transfemoral Bone-Anchored Limbs: An International Delphi Method Study," *Archives of Physical Medicine & Rehabilitation*, 2025. [Links](#).
3. **Earley EJ**, Ramne M, Wessberg J. "Unified Measures Quantifying Intensity and Similarity of Pain and Somatosensory Percepts," *Journal of Neurophysiology*, 2025. [Links](#).
4. **Earley EJ**, Kristoffersen MB, Ortiz-Catalan M. "Comparing implantable epimysial and intramuscular electrodes for prosthetic control," *Frontiers in Neuroscience*, 2025. [Links](#).
5. **Earley EJ**, Milius D, Stevens PM. "Prosthetic Component Selection for Bone-Anchored Limbs: Why New Guidelines are Needed," *Journal of Prosthetics & Orthotics*, 2025. [Links](#).
6. **Earley EJ**, Awad ME, Milius DW, Melton DH, Gaffney BMM, Stoneback JW. "Consensus, Guidelines, and Recommendations for Prosthetic Limb Prescription: A Scoping Review," *Journal of Prosthetics & Orthotics*, 2025. [Links](#).
7. Cain J, **Earley EJ**, Harold N, Potter BK, Grover P, Thomas P, Stark G, White A. "Innovations in Amputee Care in the United States: Access, Ethics, and Equity," *Prosthesis*, 2025. [Links](#).
8. Lendaro E, Van der Sluis CK, Hermansson L, Bunketorp-Käll L, Burger H, Keesom E, Widehammar C, Muñoz-Novoa M, McGuire BE, O' Reilly P, **Earley EJ**, Iqbal S, Kristoffersen MB, Stockslius A, Gudmundson L, Hill W, Diers M, Turner K, and Ortiz-Catalan M. "Treating phantom limb pain with extended reality and machine learning," *Pain*, 2025. [Links](#).
9. Smiles AB[†], **Earley EJ**[†], Jiang N, Ortiz-Catalan M. "Sensory Feedback of Grasp Security by Direct Neural Stimulation Improves Amputee Prediction of Object Slip," *Prosthesis*, 2024. [Links](#).
10. **Earley EJ**, Zbinden J, Muñoz-Novoa M, Just F, Vasan C, Sjögren Holtz A, Emadeldin M, Kolankowska J, Davidsson B, Thesleff A, Millenaar J, Jönsson S, Cipriani C, Sassu P, Brånemark R, Ortiz-Catalan M. "Cutting edge bionics in highly impaired individuals: a case of challenges and opportunities," *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, 2024. [Links](#).
11. **Earley EJ**[†], Chan NS[†], Naber A, Mastinu E, Truong MTN, Ortiz-Catalan M. "Low-Cost, Wireless Bioelectric Signal Acquisition and Classification Platform," *IEEE Access*, 2024. [Links](#).
12. Zbinden J, **Earley EJ**, Ortiz-Catalan M. "Intuitive control of additional prosthetic joints via electro-neuromuscular constructs improves functional and disability outcomes during home use," *Journal of Neural Engineering*, 2024. [Links](#).
13. **Earley EJ**, Johnson RE, Sensinger JW, Hargrove LJ. "Wrist Speed Feedback Improves Elbow Compensation and Reaching Accuracy for Myoelectric Transradial Prosthesis Users in Hybrid Virtual Reaching Task," *Journal of Neuroengineering and Rehabilitation*, 2023. [Links](#).
14. Collu R, **Earley EJ**, Barbaro M, Ortiz-Catalan M. "Non-rectangular neurostimulation waveforms elicit varied sensation quality and perceptive fields on the hand," *Scientific Reports*, 2023. [Links](#).

15. Zbinden J, Sassu P, Mastinu E, **Earley EJ**, Muñoz-Novoa M, Brånemark R, Ortiz-Catalan M. "Improved control of a prosthetic limb by surgically recreating electro-neuromuscular constructs with implanted electrodes," *Science: Translational Medicine*, 2023. [Links](#).
16. Ortiz-Catalan M, Zbinden J, Millenaar J, D'Accolti D, Controzzi M, Clemente F, Cappello L, **Earley EJ**, Mastinu E, Kolankowska J, Muñoz-Novoa M, Jönsson S, Cipriani C, Sassu P, Brånemark R. "A highly integrated bionic hand with neural control and feedback for use in daily life," *Science Robotics*, 2023. [Links](#).
17. **Earley EJ**⁺, Bernevig A⁺, Zbinden J, Ortiz-Catalan M. "Neurostimulation Artifact Removal for Implantable Sensors Improves Signal Clarity and Decoding of Motor Volition." *Frontiers in Human Neuroscience*, 2022. [Links](#).
18. **Earley EJ**, Zbinden J, Muñoz-Novoa M, Mastinu E, Smiles A, Ortiz-Catalan M. "Competitive Motivation Increased Home Use and Improved Prosthesis Self-Perception after Cybathlon 2020 for Neuromusculoskeletal Prosthesis User," *Journal of Neuroengineering and Rehabilitation*, 2022. [Links](#).
19. Lendaro E, **Earley EJ**, Ortiz-Catalan M. "Statistical analysis plan for an international, double-blind, randomized controlled clinical trial on the use of phantom motor execution as a treatment for phantom limb pain," *Trials*, 2022. [Links](#).
20. Ahkami B, Mastinu E, **Earley EJ**, Ortiz-Catalan M. "Extra-neural signals from severed nerves enable intrinsic hand movements in transhumeral amputations," *Scientific Reports*, 2022. [Links](#).
21. **Earley EJ**, Johnson RE, Sensinger JW, Hargrove LJ. "Joint Speed Feedback Improves Myoelectric Prosthesis Adaptation after Perturbed Reaches in Non Amputees," *Scientific Reports*, 2021. [Links](#).
22. **Earley EJ**, Johnson RE, Hargrove LJ, Sensinger JW. "Joint Speed Discrimination and Augmentation for Prosthesis Feedback," *Scientific Reports*, 2018. [Links](#).
23. **Earley EJ**, Hargrove LJ, Kuiken TA. "Dual Window Pattern Recognition Classifier for Improved Partial-Hand Prosthesis Control," *Frontiers in Neuroscience*, 2016. [Links](#).

Peer-Reviewed Conference Papers

Selected for Oral Presentation

1. **Earley EJ**^{*}, Mastinu E, Ortiz-Catalan M. "Cross-Channel Impedance Measurement for Monitoring Implanted Electrodes," *IEEE Engineering in Medicine and Biology Society (EMBC)*, Glasgow, Scotland, UK. July 2022. [Links](#).
2. Musolf BM^{*}, **Earley EJ**, Muñoz-Novoa M, Ortiz-Catalan M. "Analysis and Design of a Bypass Socket for Transradial Amputations," *IEEE Engineering in Medicine and Biology Society (EMBC)*, Virtual. November 2021. [Links](#).
3. **Earley EJ**^{*}, Hargrove LJ. "Modeling Expected Reaching Error and Behaviors for Motor Adaptation," *IEEE Engineering in Medicine and Biology Society (EMBC)*, Berlin, Germany. July 2019. [Links](#).
4. **Earley EJ** and Hargrove LJ^{*}. "The Effect of Wrist Position and Hand-Grasp Pattern on Virtual Prosthesis Task Performance," *IEEE RAS/EMBS International Conference on Biomedical Robotics and Biomechatronics (BioRob)*, Singapore. June 2016. [Links](#).

Selected for Poster Presentation

5. **Earley EJ**^{*}. "Development of the Myoelectric Control Resolution Assessment: A Rapid and Device-Agnostic Outcome Measure for Upper-Limb Prostheses," *International Conference on Rehabilitation Robotics (ICORR)*, Chicago, IL, USA. May 2025. [Links](#).

6. **Earley EJ***, Piazza C, Turner KL. "A Taxonomy for Commercially Available Myoelectric Terminal Devices," *Myoelectric Controls and Upper Limb Prosthetics Symposium (MEC)*, Frederickton, NB, Canada. August 2024. [Links](#).
7. Turner KL*, Hill W, **Earley EJ**, Muñoz-Novoa M, Hermansson L, Lindner H. "Refinement of New Items in the Assessment of Capacity for Myoelectric Control for Multi-Articulating Hands," *Myoelectric Controls and Upper Limb Prosthetics Symposium (MEC)*, Frederickton, NB, Canada. August 2024. [Links](#).
8. **Earley EJ***, Ortiz-Catalan M. "Neurostimulation Perception Obeys Strength-Duration Curves and is Primarily Driven by Pulse Amplitude," *International IEEE EMBS Conference on Neural Engineering (NER)*, Baltimore, MD, USA. April 2023. [Links](#).
9. **Earley EJ***, Kaveny KJ, Johnson RE, Hargrove LJ and Sensinger JW. "Joint-based velocity feedback to virtual limb dynamic perturbations," *International Conference on Rehabilitation Robotics (ICORR)*, London, GB, UK. July 2017. [Links](#).
10. **Earley EJ***, Adewuyi AA, and Hargrove LJ. "Optimizing Pattern Recognition-Based Control for Partial-Hand Prosthesis Application," *IEEE Engineering in Medicine and Biology Society (EMBC)*, Chicago, IL, USA. August 2014. [Links](#).

Non-Peer-Reviewed Abstracts and Proceedings

Selected for Oral Presentation

1. Awad ME, Melton D, **Earley E**, Gaffney BMM, Christiansen C, Stoneback JW. "Patient screening and selection for bone-anchored limb (BAL) implantation in individuals with lower extremity amputation: Eligibility criteria and risk factor analysis." *Global Collaborative Congress on Osseointegration (GCCO)*, Charlotte, NC, USA. November 2025
2. **Earley EJ***, Milius DW, Awad ME, Melton DH, Gaffney BMM, Christiansen CL, Stoneback JW. "Prosthesis Prescription Consensus for Case Patients with Transfemoral Bone-Anchored Limbs," *World Congress of the International Society for Prosthetics and Orthotics (ISPO)*, Stockholm, Sweden. June 2025. [Links](#).
3. **Earley EJ***, Milius DW, Awad ME, Melton DH, Gaffney BMM, Christiansen CL, Stoneback JW. "Prosthesis Prescription Consensus for Case Patients with Transtibial Bone-Anchored Limbs," *World Congress of the International Society for Prosthetics and Orthotics (ISPO)*, Stockholm, Sweden. June 2025. [Links](#).
4. **Earley EJ**, Awad ME, Milius DW*, Melton DH, Gaffney BMM, Stoneback JW. "An ENTREQ Meta-Synthesis of Prosthetic Limb Prescription Guidelines and Recommendations," *World Congress of the International Society for Prosthetics and Orthotics (ISPO)*, Stockholm, Sweden. June 2025. [Links](#).
5. **Earley EJ***, Shaw KG, Awad ME, Gaffney BMM, Christiansen CL, Iorio ML, Stoneback JW, Melton DH. "Phantom Limb, Residual Limb, and Bodily Pains in Persons with Unilateral Lower Limb Amputation," *World Congress of the International Society for Prosthetics and Orthotics (ISPO)*, Stockholm, Sweden. June 2025. [Links](#).
6. Awad ME, **Earley EJ**, Melton DH, Gaffney BMM, Christiansen CL, Stoneback JW*. "Patient Screening and Selection for Bone-Anchored Limb (BAL) Implantation in Individuals with Transfemoral Amputation: Evidence-Based Eligibility Criteria and Risk Factor Analysis," *World Congress of the International Society for Prosthetics and Orthotics (ISPO)*, Stockholm, Sweden. June 2025. [Links](#).

7. Turner KL*, Hill W, **Earley EJ**, Muñoz-Novoa M, Hermansson L, Lindner H. "Refining Criteria in the Assessment of Capacity for Myoelectric Control for Multi-articulating," *World Congress of the International Society for Prosthetics and Orthotics (ISPO)*, Stockholm, Sweden. June 2025. [Links](#).
8. **Earley EJ***, Awad ME, Melton DH, Hoyt C, Christiansen C, Gaffney BMM, Stoneback JW. "A Delphi Survey of OI Component Prescription Consensus," *Global Collaborative Conference on Osseointegration (GCCO)*, Charlotte, NC, USA. November 2023.
9. Just F*, Reho R, **Earley EJ**, Ortiz-Catalan M. "Neurostimulation artifact removal algorithms to improve control of a tactile feedback prosthesis," *Deutsche Gesellschaft für Biomedizinische Technik (DGBMT) Annual Conference on Biomedical Engineering*, Duisburg, Germany. September 2023. [Links](#).
10. Ahkami B*, Mastinu E, **Earley E**, Ortiz-Catalan M. "Extraneural Recordings Enable the Decoding of Intrinsic Hand Movements in Transhumeral Amputations," *World Congress of the International Society for Prosthetics and Orthotics (ISPO)*, Virtual. 2021. [Links](#).
11. **Earley EJ***, Johnson RE, Hargrove LJ and Sensinger JW. "Visual Discrimination of Biomimetic Arm Speeds," *School and Symposium on Advanced Neurorehabilitation (SSNR)*, Baiona, Spain. 2018. [Links](#).
12. **Earley E**, Kaveny K, Johnson R, Hargrove L, Sensinger J*. "Appropriate Sensory Feedback Improves Performance," *World Congress of the International Society for Prosthetics and Orthotics (ISPO)*, Cape Town, South Africa. May 2017. [Links](#).

Selected for Poster Presentation

13. **Earley EJ***, Tracy JB, Harnke B, Gaffney BMM, Christiansen CL, Stoneback JW. "A review of Osseoperception: Emergent Post-Implant Phenomenon or Distinct Sensory Experience?" *Global Collaborative Congress on Osseoperception (GCCO)*, Charlotte, NC, USA. November 2025.
14. **Earley EJ***, Milius D, Awad ME, Melton DH, Delphi Steering Committee, Christiansen CL, Gaffney BMM, Stoneback JW. "Consensus via the Delphi Method on Transfemoral Prosthesis Prescription for Bone-Anchored Limbs," *Global Collaborative Congress on Osseointegration (GCCO)*, Charlotte, NC, USA. November 2024.
15. **Earley EJ***, Milius D, Awad ME, Melton DH, Christiansen CL, Gaffney BMM, Stoneback JW. "An International Delphi Method Study to Develop Consensus on Prosthetic Prescription for Bone-Anchored Limbs," *6th Annual CU Orthopedic Research Symposium*, Denver, CO, USA. October 2024.
16. **Earley EJ***, Smiles A, Jiang N, Ortiz-Catalan M, Stoneback JW, Melton DH. "Including Prosthetic Control and Sensory Feedback in Personalized Care for Persons with Upper-Limb Loss and Limb Difference," *5th Annual CU Orthopedic Research Symposium*, Denver, CO, USA. October 2023.
17. Sanders N*, Drout M, Kohler S, Cook B, **ComSciCon Leadership Team**. "ComSciCon: The Communicating Science Workshop for Graduate Students," *American Astronomical Society Meeting #231*, Washington, DC, USA. January 2018.
18. **Earley EJ***, Kaveny KJ, Johnson RE, Hargrove LJ and Sensinger JW. "Joint-based velocity feedback improves myoelectric prosthesis performance," *Myoelectric Controls and Upper Limb Prosthetics Symposium (MEC)*, Frederickton, NB, Canada. August 2017. [Links](#).

*Publications in Preparation*Journal Articles

1. Just F[†], Reho R[†], Ortiz-Catalan M, **Earley EJ**. “Toward Simultaneous Neurostimulation and Prosthetic Control: Real-Time Filtering of Amplitude-Modulated Stimulation Artifacts from Implanted Electrode Signals.” Revision submitted.
2. **Earley EJ**, Milius DW, Awad ME, Melton DH, Ahmed K, Leijendekkers R, Potter BK, Stevens PM, Gaffney BMM, Christiansen CL, Stoneback JW. “Establishing Consensus for Prescription of Prosthetic Components for Transtibial Bone-Anchored Limbs: An International Delphi Method Study.” In review.
3. **Earley EJ**, Tracy JB, Harnke B, Gaffney BMM, Christiansen CL, Stoneback JW. “Osseoperception: Emergent Post-Implant Phenomenon or Distinct Sensory Experience? A Scoping Review.” In review.

GRANT FUNDING*Current Research Support*

- ♦ **NIH NICHD:** 1R01HD119003-01 (PI: Brecca Gaffney)
Influence of Prosthetic Foot Stiffness on Transtibial Osseointegrated Bone-Anchored Limb Outcomes
2025/08/01 – present **Role:** Collaborator

Completed Research Support

- ♦ **NRSA T32:** HD07418 (PI: Eric Perreault)
Pathophysiology & Rehabilitation of Neural Dysfunction
2016/09/01 – 2018/08/31 **Role:** Pre-Doctoral Trainee

CLINICAL TRIALS*Current Trials*

- ♦ **Balmoral Medical Company:** [NCT06134167](#) (PI: Jason W. Stoneback)
A Study to Evaluate the Safety and Effectiveness Transdermal Compress Device in Participants with Transfemoral Amputations
2024/03/26 – present **Role:** Clinical Research Manager

PROFESSIONAL MEMBERSHIPS & SERVICE*Journal Editor*

- ♦ IEEE Transactions on Medical Robotics and Bionics 2023 – present
Associate Editor, Bionic Protheses
Personally handled 20 submissions to date
- ♦ Prosthesis, MDPI 2022 – 2023
Co-Guest Editor, “Design, Control, and Biomechanics of Prosthetic Limbs.” [Special Issue link.](#)
7 submissions (personally handled 4), 4 published (personally handled 2).

Journal Referee

Over 115 verified peer reviews across 24 journals and 5 professional conferences.
Details on [Web of Science](#).

Professional Memberships

- ♦ Member, IEEE 2014 – present
- ♦ Member, IEEE Engineering in Medicine and Biology Society. 2019 – present
- ♦ Member, Robotics and Automation Society. 2022 – present
- ♦ Member, ISPO 2022 – present
- ♦ Member, AAOP 2024

*ADVISING & MENTORING**PhD Students*

- ♦ **Jan Zbinden**, Electrical Engineering (co-mentor with Max Ortiz-Catalán) 2020 – 2023
3-Degree-of-Freedom Simultaneous and Proportional Control of Prosthetic Hands. [Links](#).
- ♦ **Riccardo Collu**, Engineering & Architecture (co-mentor with Max Ortiz-Catalán & Massimo Barbaro) 2020 – 2022
Novel Waveform Shapes for Neurostimulation. [Links](#).
- ♦ **Bahareh Ahkami**, Electrical Engineering (co-mentor with Max Ortiz-Catalán) 2020 – 2021
Extra-Neural Control of Intrinsic Hand Movements in Transhumeral Amputation. [Links](#).

MS Students

- ♦ **Roberta Reho**, Robotics Engineering (co-mentor with Max Ortiz-Catalán & Fabian Just) 2022-2023
Rejection Algorithm for Modulated Neurostimulation Artifacts in iEMG signals. [Links](#).
- ♦ **Nathaly Sánchez Chan**, Biomedical Engineering (co-mentor with Max Ortiz-Catalán) 2022
ADS_BP v4 Open-Source Release. [Links](#).
- ♦ **Malin Lehander**, Biomedical Engineering (co-mentor with Mirka Buist) 2022
Improving Tactile Discrimination with Mindful Sensory Motor Training.
- ♦ **Agnes Westerlund**, Biomedical Engineering (co-mentor with Mirka Buist) 2022
Enhancing the Sense of Touch with Brain Modulation.
- ♦ **Anton Berneving**, Engineering Maths and Computer Science (co-mentor with Max Ortiz-Catalán) 2021
Neurostimulation Artifact Removal Algorithms for iEMG Prosthesis Control. [Links](#).
- ♦ **Brett Musolf**, Biomedical Engineering (co-mentor with Max Ortiz-Catalán) 2020 – 2021
Design of a Bypass Socket for Transradial Prosthesis Use. [Links](#).
- ♦ **Andrew Smiles**, Systems Design Engineering (co-mentor with Max Ortiz-Catalán & Ning Jiang) 2020 – 2021
Slip Prediction and Stimulation System for Sensorized Prosthetic Hands. [Links](#).

*Medical Students***♦ CU Orthopedics Summer Externship Program**

2025

Mentored 5 medical students leading independent research projects.

TEACHING**Chalmers University of Technology – EEMo76 – Electric Circuits and Fields 2020 – 2022**

Lecturer, Examiner

Planned and lectured for 360 undergraduate students over three years. Responsibilities included delivering 14 lectures and writing 3 examinations per year. [Course link](#).

Student course evaluation: 4.2 / 5.0

Learning objectives: analyze linear circuits using DC and AC calculation methods, perform electromagnetic field calculations based on simple geometries, and use computer-based tools to analyze simpler electrical circuits.

Course development included refining electromagnetic field module to better integrate with the circuits modules and adapting the course for digital instruction and examination during the COVID-19 pandemic.

Northwestern University – RSG Research Communication Program

2018

Graduate Assistant

Assisted running an 8-week workshop series for post docs and graduate students designed to enhance communication skills across disciplines and backgrounds. Mentored 32 students in oral presentation and video direction and editing. [Website](#).

Notre Dame University – PHIL 20632/STV 20233 – Robot Ethics

2016 – 2018

Guest Lecturer

Along with Max Shepherd, gave guest lectures titled “ProstEthics” for Prof. Don Howard’s Robot Ethics course focused on historical and current research of prosthetic limbs, and ethical considerations related to prosthetic design and transhumanism.

Nettelhorst Elementary – Get-a-Grip Program

2017

Student Mentor

Through Northwestern’s Science Club, mentored 30 elementary school students in the fundamentals of engineering design, construction, and analysis, and guided them as they developed a prosthetic device made from household items. [Website](#).

Northwestern University McCormick Graduate Leadership Council

2013 – 2017

Workshop Coordinator and Instructor

Coordinated workshops to teach introductory through advanced MATLAB and SolidWorks skills, and additional workshops on other transferable skills, for over 600 Northwestern graduate students. [Website](#).

Northwestern University – BME 307 – Quantitative Experimentation and Design

2015

Teaching Assistant

Mentored 50 biomedical engineering undergraduate students as they learned to answer questions using experimental means, and to quantify their results using statistical analysis.

LEADERSHIP

National Communicating Science Conference

2017 – present

Organizing Committee, Leadership Team, Advisory Committee

Treasurer and advisory committee member, responsible for developing fiscal procedures and managing \$300,000 in total donated funds for 30 flagship and local conferences attended by up to 1,500 graduate students. [Website](#).

Workforce for Inclusive Science

2021 - 2023

Organizing Committee

Facilitating meetings and seminars aimed at promoting and fostering equity and inclusion in academia. [Website](#).

International Conference on Phantom Limb Pain

2021

Organizing Committee

Planned, oversaw, and executed social media plan before and during the event. Moderated discussion panel during final day of the conference. [Website](#).

Chicago Communicating Science Conference

2015 – 2017

Treasurer, Lead Organizer

Organizer and treasurer for 2016 conference, lead organizer for 2017 conference. Tracked and budgeted \$24,000 in total donated funds, secured conference locations, invited keynote speakers, and oversaw conferences for 100 total attendees. [Website](#).

Northwestern University Biomedical Engineering Graduate Students Group

2013 – 2017

Co-President, Department Representative.

Oversaw academic and social events, managed \$3,500 annual budget, represented department in Graduate Leadership & Advocacy Council, and facilitated pop talks (short research summaries using jargon-free language) for annual research day. [BMEGS Website](#). [GLAC Website](#).

Colorado School of Mines Robotics Club

2008 – 2011

Treasurer, High School Mentor

Developed and managed \$26,000 annual budget. Mentored high school robotics team for three iterations of FIRST® Robotics Competition. Co-organized three FIRST® LEGO® League robotics competitions for elementary students. Co-initiated project to design, build and program self-balancing wheelchair. [Website](#).

PUBLIC OUTREACH & EDUCATION

Media Coverage

2023 – present

- ♦ “Upper-Limb Osseointegration: Connections on a Higher Plane,” O&P Edge, October 2025. [Article](#).
- ♦ “Sensations and Feelings – TSR in Upper-Limb Osteointegration,” O&P Edge, October 2025. [Article](#).
- ♦ “New Faculty Member in Department of Orthopedics Brings Expertise in Prosthetic Limb Technology,” CU Anschutz School of Medicine, July 2023. [Article](#).
- ♦ “Robothanden” documentary, Scandinavian Content Group, Sveriges Television (SVT), 2023. [Film](#).

Public Talks & Demos*2015 – present*

- ♦ “Challenges and Opportunities of Cutting Edge Bionic Prostheses,” Rocky Mountain American Society of Biomechanics Meeting, April 2024.
- ♦ Mack Clayton Invited lecture, Department of Orthopedics, University of Colorado Anschutz Medical Campus, June 2023.
- ♦ Sahlgrenska Universitetssjukhusets innovation- och teknikutbildning för läkare, 2022.
- ♦ Center for Bionics and Pain Research Annual Symposium, 2021-2022.
- ♦ PhD Thesis defense, 2019. [YouTube](#).
- ♦ “Wunderbar Together Science Slam,” Daley Plaza, Chicago, IL, 2019. [Event photos](#).
- ♦ “Neural Engineering: Designing Bionic Limbs Controlled by the Brain,” College of DuPage STEMinar Series, 2018. [YouTube](#).
- ♦ Chicago Science Festival, Illinois Science Council, 2016 – 2019.
- ♦ Museum of Science and Industry Robotics Week, 2016 – 2019.
- ♦ “How Do I Talk to my Robo-Limb?,” RSG Science Communicating Workshop, 2016. [YouTube](#).
- ♦ IEEE Engineer’s Week, 2016.
- ♦ “Adler After Dark,” Adler Planetarium, 2015 – 2018.
- ♦ Camp Neuro Chicago, 2015.

SciShow YouTube Channel*2018 – 2020*

Freelance Script Writer

Wrote easy-to-understand science video scripts for SciShow, a YouTube channel with 8 million subscribers. Topics included osseointegrated prosthetic limbs, effects of body posture, and fetal motor development, totaling 7 videos with over 1.59 million combined views. [Links](#).

Sci-Inspiration YouTube Channel*2017 – 2018*

Video Creator

Scripted, recorded, directed, and animated videos exploring scientific topics through popular media including movies, television, and video games. [Links](#).

Other Science Videos*2016 – 2020*

- ♦ “Neuromusculoskeletal Arm Prostheses”, *CBPR*, 2020. [YouTube](#).
- ♦ “STEM Connect Careers: Eric Earley.” *Discovery Education*, 2017. [Website](#).
- ♦ “Prosthetic Limbs and Motor Adaptation”, *Ready Set Go (RSG)*, 2016. [YouTube](#).

Science Writing*2017*

- ♦ The Cybathlon: The Olympics of Restoring Daily Tasks, *HELIX Magazine*, 2017. [Link](#).

ACADEMIC & TECHNICAL SKILLS

- ♦ Prosthetic sensory feedback, sensory integration, psychophysics
- ♦ Somatosensory and pain characterization and mapping
- ♦ Neuromusculoskeletal anatomy, osseointegration, implantable sensors
- ♦ Pattern recognition, classification, and machine learning algorithms
- ♦ Human motor control, motor learning and adaptation
- ♦ Statistical analysis, hypothesis testing, power analysis, linear and nonlinear mixed effects modeling
- ♦ Delphi consensus methodology, thematic analysis, systematic review, meta-synthesis
- ♦ FDA Compassionate Use/Expanded Access, Investigational Device Exemption
- ♦ FDA clinical trials, clinical research management
- ♦ International collaborations, project management, team organization, scheduling, Kanban, scrum
- ♦ Prosthetic arm standardized tests: ACMC, BBT, MCRA, MMDT, MT, PLT, RCRT, SHAP, TAC, VET
- ♦ MATLAB, Simulink, C, LabVIEW
- ♦ SolidWorks, FEA
- ♦ Adobe Illustrator, Photoshop, Premiere; Affinity Designer, Photo
- ♦ HTML5, CSS
- ♦ Native English; limited working proficiency Swedish